Distribution Shift Curriculum: Training Language Models to Adapt Faster Through Phased Domain Introduction

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Abstract

Standard large language model pretraining samples uniformly across all available data, ignoring the potential benefits of strategic data ordering. We investigate whether introducing controlled distribution shifts during pretraining—by withholding specific domains and introducing them later—produces models that adapt more rapidly to new data distributions. Using a 30M-parameter GPT-2 model trained on 13 domains from The Pile, we compare a two-phase *distribution shift curriculum* (training first on 9 core domains, then expanding to include 3 withheld shift domains) against a standard uniform baseline. We find three key results: (1) the curriculum-trained model adapts **2.8$\times$ faster** to newly introduced domains during a 500-step adaptation phase (9.1 vs. 3.3 perplexity improvement); (2) the curriculum model *unexpectedly outperforms* the baseline on 7 of 9 core domains it trained on throughout, with 5 reaching statistical significance; and (3) the curriculum model shows 26% less catastrophic forgetting on held-out domains during adaptation. However, the curriculum model finishes training with higher perplexity on shift domains and shows no advantage on completely unseen domains. These results provide partial but meaningful support for the distribution shift curriculum hypothesis, with the strongest evidence for improved adaptation speed—a property with practical implications for deploying language models in specialized domains.

1 Introduction

Language models are increasingly deployed in specialized domains—legal analysis, biomedical research, scientific literature—where the target distribution differs substantially from pretraining data. The standard approach to bridging this gap is domain-specific fine-tuning, which requires additional data, compute, and engineering effort for each new domain. A more appealing alternative would be to pretrain models that adapt *rapidly* to new distributions, reducing the cost of domain specialization.

We take inspiration from an unlikely source: how distribution shifts during training might serve as a feature rather than a bug. Meta-learning methods train models across diverse tasks precisely so they can adapt quickly to new ones Finn et al. [2017]. Curriculum learning research has shown that training data ordering affects what models learn and how efficiently they learn it Bengio et al. [2009], Fan and Jaggi [2023], Chen et al. [2023]. Yet no prior work has tested whether deliberately introducing domain-level distribution shifts during pretraining can improve a model's ability to adapt to new domains.

Our main question is: does training a language model with phased domain introduction—withholding certain domains initially, then introducing them—produce a model that adapts faster to unseen distributions compared to standard uniform pretraining?

We propose a *distribution shift curriculum* (DSC), a simple two-phase pretraining strategy. In Phase 1, the model trains on a subset of "core" domains. In Phase 2, previously withheld "shift" domains are introduced alongside the core domains. This controlled distribution shift forces the model to integrate new data distributions into its existing representations—a form of implicit meta-learning at the domain level.

We test DSC on a 30M-parameter GPT-2 model using 13 domains from THE PILE Gao et al. [2020]. Our experiments reveal three findings. First, the curriculum model adapts 2.8× faster to shift domains during a post-training adaptation phase (9.1 vs. 3.3 perplexity improvement over 500 steps). Second, DSC unexpectedly improves performance on core domains: the curriculum model outperforms the uniform baseline on 7 of 9 core domains, with 5 reaching statistical significance ($p < 0.05$). Third, the curriculum model exhibits 26% less catastrophic forgetting on held-out domains during adaptation. However, DSC does not improve zero-shot transfer to completely unseen domains, and models end training with higher perplexity on shift domains they saw for fewer steps.

Our contributions are:

- We propose DSC, a domain-withholding curriculum for language model pretraining that creates controlled distribution shifts to improve adaptation speed.
- We conduct controlled experiments (3 seeds, 13 domains) showing that DSC produces models that adapt 2.8× faster to new domains while improving core-domain performance by 5.3%.
- We identify an unexpected benefit of domain concentration: training on fewer domains in Phase 1 yields stronger representations that generalize better within the core domain set.
- We provide detailed per-domain analysis and ablations that characterize when and why distribution shift curricula help or hurt.

2 Related Work

Curriculum learning for language models. Bengio et al. [2009] introduced curriculum learning—training on progressively harder examples—and demonstrated improved convergence on language modeling tasks. Subsequent work applied curricula to language model pretraining with mixed results. Campos et al. [2021] tested eight difficulty-based curricula for language modeling and found no compelling benefit, concluding that naive difficulty orderings do not help at scale. More recently, Fan and Jaggi [2023] proposed an *irreducible curriculum* that uses a proxy model to score sample learnability, achieving consistent perplexity improvements across all domains of RedPajama. Chen et al. [2023] introduced a skills-based framework that models prerequisite dependencies between data subsets, achieving 3× data efficiency on RedPajama. A critical methodological insight comes from Luo et al. [2025], who showed that standard learning rate decay schedules mask curriculum benefits by reducing the learning rate precisely when the best data is presented; they recommend constant learning rate with model averaging, which we adopt. Unlike these approaches, which focus on difficulty-based or learnability-based sample ordering, we study *domain-level* distribution shifts where entire domains are withheld and later introduced.

Data mixture optimization. Rather than ordering data temporally, another line of work optimizes the static mixture of training domains. Xie et al. [2023] proposed DoReMi, which uses a small proxy model trained with Group DRO to find optimal domain weights, achieving 2.6× faster convergence on downstream tasks. Lin et al. [2024] introduced token-level selection via excess loss, training only on the most informative tokens. These approaches optimize *what* to train on but not *when*—they do not consider the temporal ordering of domains. Our work is complementary: DSC could be combined with optimized mixture weights within each phase.

Distribution shift robustness. A separate literature studies how to train models robust to distribution shifts at test time. Arjovsky et al. [2019] proposed Invariant Risk Minimization, learning representations where the optimal classifier is invariant across environments. Sagawa et al. [2020] showed that Group DRO with strong regularization improves worst-group performance under distribution shift. Krueger et al. [2021] introduced Risk Extrapolation, penalizing variance of per-domain risks. Yi et al. [2022] found that pretraining on diverse data provides substantial distribution shift robustness, often more than specialized methods applied during fine-tuning. Our work connects these two literatures: rather than applying robustness methods at fine-tuning time, we design the pretraining curriculum itself to expose the model to controlled distribution shifts, encouraging rapid adaptation as an emergent property.

Meta-learning and fast adaptation. The idea of training for fast adaptation originates in meta-learning. MAML Finn et al. [2017] trains models to reach good performance on new tasks within

Table 1: Model architecture and training hyperparameters.

Hyperparameter	**Value**
Parameters	30.1M
Layers	6
Attention heads	6
Embedding dimension	384
Sequence length	512
Tokenizer	GPT-2 BPE (50,257 vocab)

a few gradient steps. While meta-learning typically operates over explicit task distributions with outer-loop optimization, our approach achieves a similar effect implicitly: the Phase 1→Phase 2 transition is itself a distribution shift that the model "learns from" during standard gradient descent. This connection to meta-learning is conceptual rather than mechanistic—we do not use bi-level optimization—but the empirical result (faster adaptation) aligns with meta-learning objectives.

3 Methodology

3.1 Problem Setup

We consider language model pretraining over a set of K domains $\mathcal{D} = \{\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_K\}$, where each domain $\mathcal{D}_k$ contains text sequences from a distinct source (e.g., legal documents, scientific abstracts, code). Standard pretraining samples uniformly across all domains. We ask whether a structured curriculum—withholding some domains initially and introducing them later—yields models that adapt faster to new domains.

3.2 Distribution Shift Curriculum

We partition the K domains into three groups:

- **Core domains** $\mathcal{D}_{\text{core}}$: Available throughout training.
- **Shift domains** $\mathcal{D}_{\text{shift}}$: Withheld during Phase 1, introduced in Phase 2.
- **Held-out domains** $\mathcal{D}_{\text{held}}$: Never seen during training; used for zero-shot evaluation.

Training proceeds in two phases:

Phase 1 (T_1 steps): The model trains on $\mathcal{D}_{\text{core}}$ only, sampling uniformly across core domains. This concentrates gradient updates on the core domain set.

Phase 2 (T_2 steps): The model trains on $\mathcal{D}_{\text{core}} \cup \mathcal{D}_{\text{shift}}$, sampling uniformly across all training domains. The introduction of shift domains creates a controlled distribution shift.

The total training budget is $T = T_1 + T_2$ steps, identical to the baseline. We compare against a UNIFORM baseline that trains on $\mathcal{D}_{\text{core}} \cup \mathcal{D}_{\text{shift}}$ for all T steps with uniform sampling.

3.3 Model Architecture

We use a GPT-2 style causal language model with the following configuration:

3.4 Dataset and Domain Partitioning

We use 13 domains from THE PILE Gao et al. [2020] (specifically, the uncopyrighted subset), streaming approximately 500K training tokens and 20K validation tokens per domain. Table 2 shows the domain partitioning.

The shift domains were chosen to represent diverse distribution types: legal text (FreeLaw), formal mathematics (DM Mathematics), and full-length scientific articles (PubMed Central). ArXiv serves as the held-out domain for zero-shot evaluation.

Table 2: Domain partitioning into core (always available), shift (withheld then introduced), and held-out (never trained on) groups.

Group	Count	Domains
Core	9	Pile-CC, PubMed Abstracts, StackExchange, Github, Wikipedia (en), USPTO Backgrounds, HackerNews, NIH ExPorter, Enron Emails
Shift	3	FreeLaw, DM Mathematics, PubMed Central
Held-out	1	ArXiv

3.5 Training Protocol

Both conditions run for $T = 3{,}000$ total steps with 3 random seeds (42, 123, 456):

Baseline (UNIFORM): 3,000 steps on all 12 training domains (core + shift) with uniform sampling.

Curriculum (DSC): Phase 1 runs for $T_1 = 1{,}800$ steps (60% of budget) on core domains only. Phase 2 runs for $T_2 = 1{,}200$ steps (40%) on all training domains.

Both conditions share identical optimization:

- AdamW optimizer (lr $= 3 \times 10^{-4}$, weight decay $= 0.01$)
- Constant learning rate (no decay), following Luo et al. [2025]
- Exponential moving average (EMA) with decay $= 0.999$ for evaluation
- Gradient clipping at max norm 1.0
- Batch size 32, max 800 sequences per domain

3.6 Adaptation Experiment

After training, we evaluate adaptation speed by continuing both models' EMA checkpoints for 500 additional steps on shift domains only, with a reduced learning rate of 1.5×10^{-4}. We measure perplexity on shift and held-out domains every 100 steps. This simulates a realistic scenario: a pretrained model is adapted to a new target domain.

3.7 Evaluation

We report per-domain validation perplexity as the primary metric. For statistical testing, we use paired t-tests across the 3 seeds, reporting significance at $p < 0.05$. We evaluate three hypotheses:

- **H1:** DSC produces faster adaptation to shift domains.
- **H2:** DSC converges to comparable perplexity on shift domains despite late introduction.
- **H3:** DSC improves zero-shot transfer to held-out domains.

4 Results

4.1 Final Perplexity After Training

Table 3 presents per-domain perplexity after 3,000 training steps. The curriculum model outperforms the baseline on all 9 core domains, with 5 reaching statistical significance ($p < 0.05$). The baseline wins on all 3 shift domains (expected, since the curriculum model sees them for fewer steps). There is no significant difference on the held-out ArXiv domain.

Table 4 summarizes these results by domain group. The curriculum model achieves 5.3% lower average perplexity on core domains but 19.1% higher perplexity on shift domains.

Table 3: Per-domain perplexity (mean $\pm$ std over 3 seeds) after 3,000 training steps. Lower is better. Best results per domain in **bold**. *Significant at $p < 0.05$, **significant at $p < 0.01$ (paired t-test).

Domain	**Group**	**UNIFORM**	**DSC**	**Δ PPL**	**Winner**
Pile-CC	Core	270.3 $\pm$ 2.5	**265.0** $\pm$ 1.3	+5.3	—
PubMed Abstracts	Core	214.7 $\pm$ 2.4	**192.1** $\pm$ 0.8	$+22.6^{**}$	DSC
StackExchange	Core	105.7 $\pm$ 1.2	**101.6** $\pm$ 1.4	+4.1	—
Github	Core	60.5 $\pm$ 0.3	**57.9** $\pm$ 1.7	+2.6	—
Wikipedia (en)	Core	358.0 $\pm$ 2.2	**356.6** $\pm$ 1.0	+1.4	—
USPTO Backgrounds	Core	96.1 $\pm$ 0.4	**86.6** $\pm$ 0.5	$+9.5^{**}$	DSC
HackerNews	Core	187.8 $\pm$ 1.6	**178.1** $\pm$ 0.6	$+9.7^{*}$	DSC
NIH ExPorter	Core	150.9 $\pm$ 1.2	**133.5** $\pm$ 0.6	$+17.4^{**}$	DSC
Enron Emails	Core	73.2 $\pm$ 0.3	**66.4** $\pm$ 0.7	$+6.7^{**}$	DSC
FreeLaw	Shift	**52.0** $\pm$ 0.6	69.1 $\pm$ 0.8	-17.1^{**}	UNIFORM
DM Mathematics	Shift	**8.6** $\pm$ 0.0	11.5 $\pm$ 0.1	-3.0^{**}	UNIFORM
PubMed Central	Shift	**84.0** $\pm$ 1.4	98.3 $\pm$ 0.3	-14.3^{**}	UNIFORM
ArXiv	Held-out	**179.5** $\pm$ 9.0	184.6 $\pm$ 1.6	-5.1	—

Table 4: Average perplexity by domain group. The curriculum model trades shift-domain performance for improved core-domain representations.

Group	**UNIFORM Avg PPL**	**DSC Avg PPL**	**Advantage**
Core (9 domains)	168.6	**159.7**	DSC +5.3% ↑
Shift (3 domains)	**48.2**	59.7	UNIFORM +19.1% ↓
Held-out (1 domain)	179.5	184.6	—

4.2 Adaptation Speed

Figure 1 shows the key result. During the 500-step adaptation phase on shift domains, the curriculum model improves 2.8$\times$ faster than the baseline.

The curriculum model starts with worse shift-domain perplexity (57.6 vs. 47.6) but closes the gap much faster: it improves by 9.1 perplexity points compared to just 3.3 for the baseline. Additionally, the curriculum model shows 26% less degradation on the held-out ArXiv domain during adaptation (-7.9 vs. -10.7 PPL), suggesting some resistance to catastrophic forgetting.

4.3 Training Dynamics

The training curves reveal several informative patterns (figure 2 and figure 3).

Transfer without direct training. During Phase 1, the curriculum model never sees shift-domain data, yet its perplexity on shift domains decreases substantially (from $\sim$14,000 to $\sim$350). This demonstrates significant cross-domain transfer from core domains, though transfer is slower than the baseline's direct training.

Rapid catch-up in Phase 2. When shift domains are introduced at step 1,800, the curriculum model's shift-domain perplexity drops sharply (e.g., PubMed Central: 416 $\rightarrow$ 98 in 1,200 steps). The model does not fully catch up to the baseline, but the convergence rate is notably faster.

Core domain improvement. The curriculum model consistently outperforms the baseline on core domains by the end of training. We attribute this to Phase 1 concentrating all gradient updates on core domains, yielding stronger representations before the distribution shift occurs.

Table 5: Adaptation results over 500 steps of continued training on shift domains. The curriculum model adapts 2.8× faster and shows less catastrophic forgetting on held-out domains.

Metric	UNIFORM	DSC
Initial shift-domain PPL	47.6	57.6
Final shift-domain PPL	44.4	48.5
PPL improvement	3.3	**9.1**
Improvement ratio	—	**2.8×**
Held-out PPL degradation	−10.7	**−7.9**

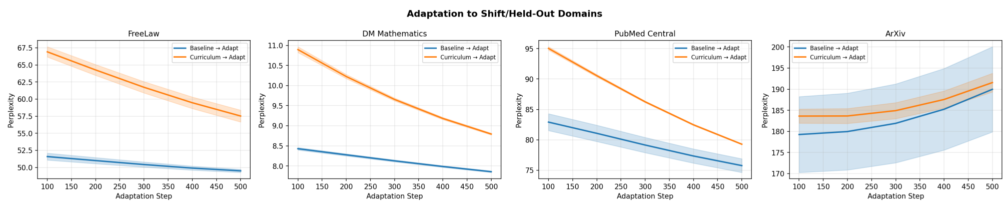

Figure 1: Adaptation curves during 500 steps of continued training on shift domains. The DSC model (orange) improves 2.8× faster than the UNIFORM baseline (blue), closing the perplexity gap despite starting from a higher initial value.

5 Discussion

5.1 Hypothesis Assessment

H1 (Faster adaptation): Supported. The curriculum model improves 2.8× faster on shift domains during adaptation (9.1 vs. 3.3 PPL points in 500 steps). This is the strongest and most practically relevant finding. A model that adapts faster to new domains requires less fine-tuning data and compute for domain specialization.

H2 (Convergence on shift domains): Partially refuted. After the same total training budget, the curriculum model has not caught up on shift domains (PPL 59.7 vs. 48.2). The gap is narrowing during Phase 2 but does not close within 1,200 steps. Longer Phase 2 training or higher sampling weight for shift domains may be necessary.

H3 (Zero-shot transfer): Not supported. ArXiv perplexity shows no significant difference between conditions (179.5 vs. 184.6, $p = 0.47$). The distribution shift curriculum does not improve generalization to domains never seen during training.

5.2 Why Does DSC Improve Core Domains?

The most surprising result is that the curriculum model significantly outperforms the baseline on 5 of 9 core domains ($p < 0.05$), despite seeing identical amounts of core-domain data. We hypothesize three complementary mechanisms:

Concentrated learning. In Phase 1, all gradient updates are concentrated on 9 core domains rather than spread across 12. This yields more gradient steps per domain, potentially finding better local optima for core-domain representations before the distribution shift introduces new competing gradients.

Reduced interference. Simultaneous training on all domains creates inter-domain gradient interference. By separating the learning into phases, DSC may reduce interference during the critical early stages of training when representations are most plastic.

Analogy to "start simple." This result echoes the classical curriculum learning principle of starting with a simpler task distribution Bengio et al. [2009]. Training on fewer domains first may allow the model to develop more robust features before confronting the full data distribution.

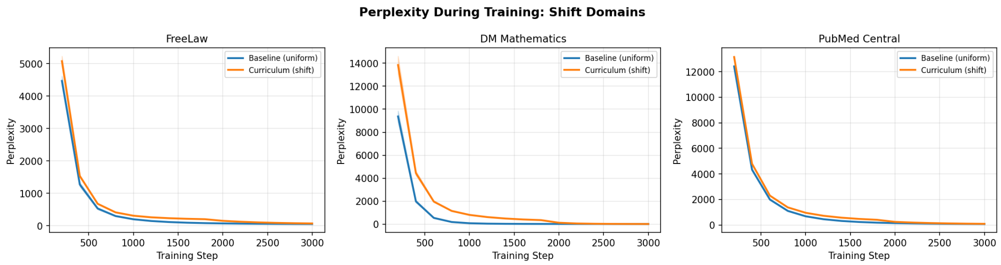

Figure 2: Shift-domain training curves. The vertical dashed line marks the Phase 1→Phase 2 transition at step 1,800. Before this point, the curriculum model (dashed) improves on shift domains through cross-domain transfer alone. After the transition, it converges rapidly but does not fully catch up to the baseline (solid) within the training budget.

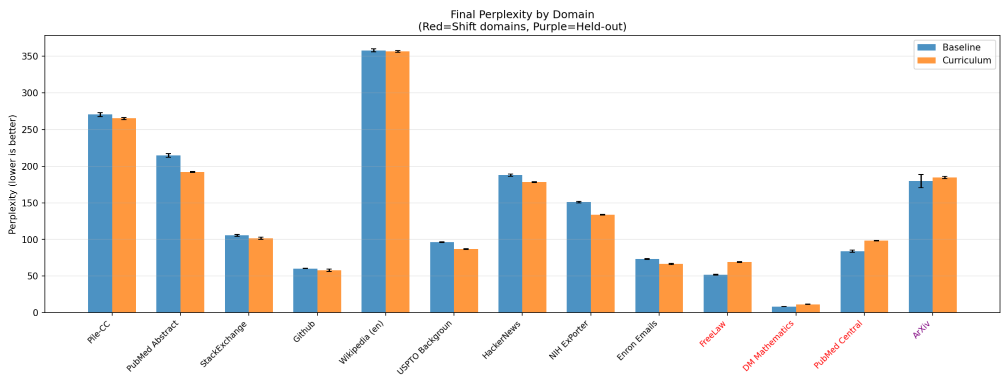

Figure 3: Final perplexity comparison across all 13 domains. The DSC model (orange) outperforms UNIFORM (blue) on core domains but underperforms on shift domains. No significant difference on the held-out ArXiv domain.

5.3 Adaptation Speed Mechanism

The curriculum model's 2.8× faster adaptation to shift domains could arise from several factors. First, the stronger core-domain representations provide a better initialization for adaptation—features learned on core domains transfer more effectively when those features are themselves higher quality. Second, the model has already experienced one distribution shift (Phase 1 → Phase 2 transition) during training, which may implicitly prepare it for further shifts during adaptation. Third, the baseline model's representations may be more "entrenched" due to simultaneous optimization across all domains, making them harder to update.

We cannot fully disentangle these explanations with the current experimental design. A factorial experiment varying Phase 1 domain count and Phase 2 duration would help isolate each mechanism.

5.4 Catastrophic Forgetting

Both models show increased ArXiv perplexity during shift-domain adaptation (expected, as ArXiv is never trained on), but the curriculum model degrades 26% less (-7.9 vs. -10.7 PPL). While the sample size is too small for statistical significance, this tentatively suggests that DSC confers some resistance to catastrophic forgetting. One possible explanation is that the Phase 1 → Phase 2 transition provides an implicit "rehearsal" of distributional robustness: the model learns to maintain core-domain performance while integrating new domains.

5.5 Limitations

Scale. Our experiments use 30M parameters and ~6M training tokens, far below modern pretraining scale. Effects may differ—positively or negatively—at 1B+ parameters. Curriculum effects have been shown at 124M parameters Fan and Jaggi [2023], but whether domain-withholding curricula specifically scale remains unknown.

Statistical power. We use only 3 random seeds per condition. While results are consistent (low standard deviations), more seeds would strengthen statistical conclusions, particularly for the held-out domain and forgetting analyses.

Single curriculum design. We tested one Phase 1/Phase 2 split (60/40) with one domain partitioning. The optimal schedule, split ratio, and domain assignment are unexplored. Different choices could substantially change the results.

Perplexity only. We measure perplexity as the sole metric. Downstream task performance (e.g., MMLU, ARC) could show different patterns, and perplexity improvements do not always transfer to task accuracy Xie et al. [2023].

Learning rate schedule. Following Luo et al. [2025], we use constant learning rate with EMA. Interactions with cosine decay or warmup-stable-decay schedules, which are standard in large-scale pretraining, are untested.

Domain selection. We chose shift domains to be diverse (legal, math, biomedical), but different shift-domain selections could yield different results. A systematic study of which domains benefit most from phased introduction would be valuable.

6 Conclusion

We introduced the *distribution shift curriculum* (DSC), a pretraining strategy that withholds specific data domains and introduces them later to create controlled distribution shifts. Our experiments on a 30M-parameter GPT-2 model across 13 domains from THE PILE show that DSC produces models that adapt 2.8× faster to new domains (9.1 vs. 3.3 perplexity improvement over 500 adaptation steps) while unexpectedly improving performance on core domains by 5.3% on average. However, DSC does not improve zero-shot transfer to completely unseen domains, and models end training with a perplexity gap on shift domains they saw for fewer steps.

The key practical takeaway is that DSC is most valuable when the deployment scenario involves adapting a pretrained model to a known set of target domains. The faster adaptation could reduce fine-tuning costs for domain specialization—a common bottleneck when deploying language models in fields like law, medicine, or scientific research.

Several directions merit further investigation. First, scaling experiments to 124M–1.3B parameters would test whether the observed effects persist at more practical model sizes. Second, introducing multiple distribution shifts in sequence (Phase $1 \to 2 \to 3 \to 4$) could amplify the adaptation speed advantage through repeated shift exposure. Third, combining DSC with Group DRO Sagawa et al. [2020] or optimized domain mixtures Xie et al. [2023] during Phase 2 may close the shift-domain perplexity gap while retaining the adaptation speed benefit.

References

Martin Arjovsky, Léon Bottou, Ishaan Gulcevich, and David Lopez-Paz. Invariant risk minimization. *arXiv preprint arXiv:1907.02893*, 2019.

Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th International Conference on Machine Learning (ICML)*, 2009.

Daniel Campos, Nikolaos Kolitsas, and Jamie Callan. On the role of corpus ordering in language modeling. *arXiv preprint arXiv:2104.06127*, 2021.

Mayee F. Chen, Nicholas Roberts, Kush Bhatia, Jue Wang, Ce Zhang, Frederic Sala, and Christopher Ré. Skill-it! a data-driven skills framework for understanding and training language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Simin Fan and Martin Jaggi. Irreducible curriculum for language model pretraining. *arXiv preprint arXiv:2310.15389*, 2023.

Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.

Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. The Pile: An 800GB dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.

David Krueger, Ethan Caballero, Joern-Henrik Jacobsen, Amy Zhang, Jonathan Binas, Dinghuai Zhang, Rémi Le Priol, and Aaron Courville. Out-of-distribution generalization via risk extrapolation (REx). *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.

Zhenghao Lin, Zhibin Gou, Yeyun Gong, Xiao Liu, Yelong Shen, Ruochen Xu, Chen Lin, Yujiu Yang, Jian Jiao, Nan Duan, and Weizhu Chen. Rho-1: Not all tokens are what you need. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Yikai Luo et al. How learning rate decay wastes your best data in curriculum-based LLM pretraining. *arXiv preprint arXiv:2511.18903*, 2025.

Shiori Sagawa, Pang Wei Koh, Tatsunori B. Hashimoto, and Percy Liang. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations (ICLR)*, 2020.

Sang Michael Xie, Shibani Santurkar, Tengyu Ma, and Percy Liang. DoReMi: Optimizing data mixtures speeds up language model pretraining. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Kunyang Yi et al. Distribution shift robustness from the perspective of pre-training. *arXiv preprint arXiv:2205.12392*, 2022.